

The Drain of Scientific Publishing

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Abstract: The domination of scientific publishing in the Global North by major commercial publishers is harmful to science; we need the most powerful members of the research community – funders, governments and Universities – to lead the drive to re-communalise publishing to serve science not the market.

Introduction

Scholarly journals disseminate knowledge that advances our understanding of humanity, life and the universe. But they serve other purposes. They provide recognition and influence for researchers, their institutions and funders. Journals also earn revenue for commercial publishers, turning prestige into profit. These three purposes – knowledge, prestige and profit – are now poorly aligned.

In this article, we show that the relationships that have developed between researchers, their funders and commercial publishers are draining the research system, despite (sometimes even, due to) recent efforts to embrace Open Access publishing models. The drain is four-fold, depriving the research system of Money, Time, Trust and Control. In some languages, disciplines and regions, different publishing practices provide welcome alternatives. But as most researchers rely on publication in commercial journals to secure status and climb the prestige hierarchies, the drain continues to be at work.

In their early days, journals served small, dedicated communities of readers and often survived on philanthropy, altruism or institutional support (1). However, since the 1950s publications have become key tokens in the increasingly fierce competition for prestige. The number of publications worldwide increased exponentially. During the same period, commercial publishers took over from older non-profits as the dominant forces in what had, by the late twentieth century, become a highly profitable industry.

There is a long history of governments and other funders supporting research journals. Examples include the UK government (from 1895), the Rockefeller Foundation (from the 1930s), the Nuffield Foundation (in the 1950s) and the US federal government (from the 1960s). But historically, funders were helping non-profit publishers break even, rather than boosting the profits of private enterprise. The success of the profit-making model of scientific publishing in the late twentieth century enabled funders to step back from supporting journals.

The Open Access movement has brought funders back to the table. They are not yet wielding their influence as much as they could, but they need to. Commercial publishers have managed to monetize funder mandates for Open Access (2). Author publication fees have become new revenue streams. Rather than democratizing scientific publishing, Open Access has helped commercial publishers generate more profits. More stringent reforms are required to tackle the misaligned drivers of scientific publishing.

Below we describe the drain in more detail and its harms. We argue that publishers' interests have successfully compromised attempts to stop it. We then explain how it may be more effectively addressed. These reforms require research communities to reclaim journals in order to address the drain and Universities, governments and funders to exert their full influence to support them to do so.

1. The four-fold drain

1.1 Money

Currently, academic publishing is dominated by profit-oriented, multinational companies for whom scientific knowledge is a commodity to be sold back to the academic community who created it. The dominant four are Elsevier, Springer Nature, Wiley and Taylor & Francis, which collectively generated over US\$7.1 billion in revenue from journal publishing in 2024 alone, and over US\$14 billion in profits between 2019 and 2024 (Table 1A). Their profit

margins have always been over 30% in the last five years, and for the largest publisher (Elsevier) always over 37%.

Against many comparators, across many sectors, scientific publishing is one of the most consistently profitable industries (Table S1). These financial arrangements make a substantial difference to science budgets. In 2024, 46% of Elsevier revenues and 53% of Taylor & Francis revenues were generated in North America, charging North American researchers over US\$2.27 billion. The National Science Foundation budget that year was \$ 9.1 billion and that of NSERC in Canada 1.1 billion.

Table 1. Scientific Publishing Profits 2019-2024 and APCs 2019-2023

A • Annual revenues and profits, millions USD, selected publishers

	2019	2020	2021	2022	2023	2024
Elsevier						
Revenue	3368	3456	3644	3598	3809	3899
Profit	1254	1311	1377	1361	1449	1498
Profit margin	37%	38%	38%	38%	38%	38%
Springer Nature						
Revenue	1425	1435	1598	1383	1484	1530
Profit	407	393	455	446	473	489
Profit margin	29% [†]	27% [†]	28% [†]	32% [†]	32%	32%
Taylor & Francis						
Revenue	715	714	750	734	770	892
Profit	279	275	281	258	271	327
Profit margin	39%	39%	37%	35%	35%	37%
Wiley						
Revenue	939	949	1015	1111	1080	1043
Profit	322	335	357	390	377	331
Profit margin	34%	35%	35%	35%	35%	32%

[†] Springer Nature profits for 2019-22 estimated based on 2023-24 data.

Source: Annual financial reports of publishers (see Materials and Methods).

B • Estimated APCs, millions USD, selected publishers

	2019	2020	2021	2022	2023
Elsevier	137	187	322	403	583
Frontiers	89	132	230	331	258
MDPI	168	306	501	657	682
Springer Nature	251	329	429	478	547
Wiley	101	167	182	255	429

Source: (3)

Researchers tend to care little about how much is being paid to publishers or feel powerless to affect change. As a result, publishing in this prestige system demonstrates limited price sensitivity. Barriers to entry are still high and the market has become more concentrated. In the past few decades, through mergers and acquisitions, an oligopoly emerged (4).

During the transition to digital publishing in the early 2000s, publishers offered libraries so-called “Big Deals” or bundled subscription packages that locked libraries into multi-year, opaque contracts governed by Non-Disclosure Agreements (5). Under the Open Access banner, these packages have evolved into “Read-and-Publish Agreements”, folding subscriptions and article processing charges (APCs) into a single, similarly opaque invoice for research institutions with little choice but to buy or lose access to scientific literature.

APCs have exacerbated the distortions of commercial publishing. Whereas the Open Access movement aimed to make knowledge freely accessible, publishers found ways to shift paywalls from readers to authors. In some countries in the majority world, where public funds are unavailable, researchers will sometimes meet these costs personally from meagre salaries. APCs now form an increasing part of their lucrative business models (2). Between 2019-2023, Haustein and colleagues (3) estimated that the top publishers amassed close to \$US 9 billion from APC revenues (Table 1B). In today’s digital environment, commercial publishers thrive on subscription bundles, APC revenues, and selling the data their publishing work provides (6).

The situation is not globally uniform. In Europe, many journals run by learned societies or subject associations are now run in partnership with - or legally transferred to - commercial publishers. However, in Latin America, the majority of journals are still sustained by public universities. As a result, thousands of autonomous, community-led, diamond Open Access journals still thrive outside the publishing oligopoly (7, 8, S1). But beyond their sphere, money flowing to commercial publishers drains research resources. Worse still, the incentives publishers face to sell more papers and glean the data they provide drive the other three aspects of the drain we describe.

1.2 Time

The number of papers published each year is growing faster than the scientific workforce, with the number of papers per researcher almost doubling between 1996 and 2022 (Figure 1A). This reflects the fact that publishers’ commercial desire to publish (sell) more material has aligned well with the competitive prestige culture in which publications help secure jobs, grants, promotions, and awards. To the extent that this growth is driven by a pressure for profit, rather than scholarly imperatives, it distorts the way researchers spend their time.

The publishing system depends on unpaid reviewer labour, estimated to be over 130 million unpaid hours annually in 2020 alone (9). Researchers have complained about the demands of peer-review for decades, but the scale of the problem is now worse, with editors reporting widespread difficulties recruiting reviewers. The growth in publications involves not only the authors’ time, but that of academic editors and reviewers who are dealing with so many review demands.

Even more seriously, the imperative to produce ever more articles reshapes the nature of scientific inquiry. Evidence across multiple fields shows that more papers result in ‘ossification’, not new ideas (10). It may seem paradoxical that more papers can slow progress until one considers how it affects researchers’ time. While rewards remain tied to volume, prestige, and impact of publications, researchers will be nudged away from riskier, local, interdisciplinary, and long-term work. The result is a treadmill of constant activity with